What LLMs Can't See About Gender

Anonymized

This paper brings theorizing from Trans Studies to bear on recent Natural Language Processing (NLP) methods for detecting and evaluating trans bias in Large Language Models (LLMs). Drawing in particular on Trans Studies' critique of gender identification, and its emphasis on gender as a visible, expressive, and social phenomenon, the paper examines how treating gender primarily as an internal identity limits what bias detection systems are able to perceive. According to this critique, systems that conceptualize gender as internal render dominant identities invisible while obscuring external expression and presentation which are most frequently targeted by discrimination. Through close analysis of three recent frameworks for studying gender bias that address trans identities—the TIDEs, EQUITBL, and TANGO frameworks [31, 15, 33]—the paper considers to what extent existing techniques for evaluating bias respond to this the concept of gender as external and expressive. It concludes by outlining directions for future research that integrate theoretical insights from Trans Studies to better capture the material and relational dimensions of gender bias in language technologies.

Note: this paper uses the term "trans" to refer to all identities that do not fall under cisgender male and cisgender female identities, including nonbinary, transfeminine, and transmasculine identities, among others.

CCS CONCEPTS • **Computing Methodologies** • Artificial Intelligence • Natural language processing

Additional Keywords and Phrases: NLP, Gender bias, Transgender, Trans, Trans Studies, Bias Evaluation and Mitigation

ACM Reference Format:
First Author's Name, Initials, and Last Name, Second Author's Name, Initials, and Last Name, and Third Author's Name, Initials, and Last Name. 2018. The Title of the Paper: ACM Conference Proceedings Manuscript Submission Template: This is the subtitle of the paper, this document both explains and embodies the submission format for authors using Word. In Woodstock '18: ACM Symposium on Neural Gaze Detection, June 03–05, 2018, Woodstock, NY. ACM, New York, NY, USA, 10 pages. NOTE: This block will be automatically generated when manuscripts are processed after acceptance.

1 INTRODUCTION

There is a global urgency to define gender. In various countries around the world, gender designations like "transgender" and "nonbinary" are prompting backlash that has resulted in the retraction of hard-gained rights for LGBTQ+ groups, such as the recent ban on gender-affirming care for minors in the USA [28]. This backlash takes what it terms "gender ideology", or nontraditional views of gender that are not based on biological sex, as a rallying cry for political priorities that align with patriarchal and xenophobic logics, such as anti-immigrant discrimination [8, 25, 36].

In response to attacks on so-called gender ideology, some progressive groups have doubled down on a definition of gender that is based on internal identification rather than biological reality [16]. Drawing from foundational theories of gender, such as Judith Butler's [7] theory of Gender Performativity (discussed in more detail below), which breaks down the rigid and causal relationship between sex and gender, this perspective leaves room for individuals to

identify according to their personal perception and feeling, which may or may not correspond to their assigned sex at birth.

In contrast to the progressive view, the academic field of Trans Studies has been exploring paradigms that describe and represent non-normative forms of gender according to its expressive characteristics. One perspective from this field, called "trans materialism", is based on the notion that gender operates externally, and ought to be evident in behavior, social relations, and appearance [1]. Its logic, simply put, states that if gender is conceived as an external phenomenon, it will be easier to make visible those who are non-conforming to gender norms, and therefore most marginalized.

Across the disciplinary divide, NLP has been developing various methods for studying gender bias in language forms. Over the past several years, such methods have ranged from Word Embedding approaches [6], to Coreference Resolution tasks [35], and more recently, Language Generation methods that include prompting [27]. Most recently, a growing number of these methods address queer and trans bias in particular [19, 33, 31, 15]. Through the analysis of concrete linguistic features, these bias detection and mitigation techniques aim to make explicit the ways that language forms encode harmful perceptions of gender.

This paper explores how theorizing from Trans Studies might be applied to current methods in NLP, focusing on three recent frameworks for detecting trans bias in language: TIDEs, EQUITBL, and TANGO [31, 15, 33]. A close reading of each of these three frameworks examines in detail how gender is operationalized and how language makes gender forms visible. The following discussion then draws out potential sites of engagement with Trans Studies theory. It argues that two of these frameworks, TIDEs and EQUITBL, despite contrasting approaches (one is quantitative and the other qualitative) significantly benefit from addressing gender as a behavioral and social phenomenon, as theorized in Trans Studies. Conversely, the third approach, TANGO, is found to be limited by its focus on gender as an exclusively internal concept. The discussion thus illustrates how Trans Studies' theorizing on expressive and behavioral aspects of gender identity has direct application to techniques for marking up and coding gender in NLP. As such, this paper contributes to the small but increasing momentum around deconstructing "gender" as a concept within NLP by drawing from knowledge domains beyond computer science [23, 9].

A central part of this paper's motivation is its analytical method, which it demonstrates in practice in its analysis of the three bias detection and mitigation techniques. This paper posits that the sustained attention to individual samples of evidence, called "close reading" in Humanities disciplines, can draw out significant alignments between diverse methodologies. It therefore contributes to work from this venue that calls for interdisciplinary theorizing in NLP [43, 4].

2 GENDER BIAS IN NLP

Much of NLP research on gender bias is driven by various assumptions about various concepts like "bias" and "gender equality". The trend is powerfully articulated by Blodgett et al.'s [5] landmark paper in NLP bias research, which points out that much of this research does not define underlying terms like "bias", as well as associated harms. This paper argues that when the motivation for analyzing bias is not explicit, then the techniques used to address bias appear arbitrary. For example, the operationalization of terms like "discriminate," "systematic bias", and "social injustice" in ways that are not intelligible run the risk of being improperly matched to detection and mitigation techniques.

Additionally, techniques for detecting and mitigating gender bias often assume that gender equality means that gender terms ought to be interchangeable (i.e., switching "male" for "female" in text generation tasks). Word embedding [6] and Coreference Resolution tasks [35], for example, are based on the idea that each of the genders ought

to have equal associations with all attributes (i.e. that a male gendered term ought to be as equally associated as a female one to attributes such as "babysitter" or "computer programmer") [6]. Meanwhile, other research [21, 23, 9] points out that such frameworks do not consider how gender terms contain deeper associations with social norms. For example, [21] find that after debiasing, words like "nurse," while no longer associated with "explicitly marked feminine words," maintains its association to "socially-marked feminine words," like "receptionist" and "teacher".

One major assumption that guides gender bias research is the gender binary itself, the assumption of gender as male or female [6, 22, 21]. This tendency has been strongly critiqued in work that emphasizes LGBTIQIA+ identities and which, increasingly, centers these communities in the creation of bias detection and mitigation techniques [19, 14, 31, 33]. From this research, one major trend is relevant to this project: the interrogation of how key terms like "gender" are being defined and operationalized. [23, 9], for example, draw from the insights of seminal gender theorist, Judith Butler, in order to conceptualize gender as a social construction. While [23] uses this construction to show the harmful ways that sex and gender have been collapsed in language model outputs, [9] uses it to base a critique of dominant bias detection and mitigation techniques.

3 DEFINING GENDER

The next subsections consider how gender has been conceptualized between popular discourses and the academic field of Trans Studies.

3.1 Gender Ideology in Popular Discourse

The term "gender ideology", which the Southern Poverty Law Center classifies as a derogatory term, refers to the view that trans identities are internal and therefore false belief systems [37]. In various countries around the world, gender ideology has risen to a national issue, serving to unite various political aims into what are mostly alt-right agendas. In the European continent, conservative groups have used gender ideology to prop up other forms of social discrimination, such as that against migrants. In Italy, for example, discourses of gender ideology have been linked to white nationalism, whereby "'gender' can be used to 'stick' 'migrant' and queer subjects together, characterizing both as threats to the sexual order of Europe" [26]. Meanwhile, on the African continent, similar arguments are used toward politically opposite messaging, in anti-colonial discourses, where gender ideology is framed as a dangerous, Western import that threatens traditional gender roles. In Uganda, in particular, resistance against gender ideology has prompted a wave of anti-LGBT+ legislation, with one law penalizing what it calls "aggravated homosexuality" with the death penalty [2]. Across these various political contexts, ranging from anti-immigrant to anti-West framings, gender ideology arguments tend to center on patriarchal arguments about the traditional sexual order.

In the USA, concerns over gender ideology have prompted a dramatic increase in legislation and policies which limits the rights of trans people. Over the last five years, the number of proposed anti-trans bills each year has surpassed the last, with 2025 seeing over 1000 such bills being proposed in legislative houses across the country [41]. Additionally, last year, the White House released a series of Executive Orders with titles that frame trans discrimination as the protection of women: "Defending Women From Gender Ideology Extremism And Restoring Biological Truth To The Federal Government" and "Keeping Men Out of Women's Sports" the [39, 40]. These orders deploy a common, right-wing talking point that so-called "biological" women need explicit protection from trans women (this argumentation also appears in TERF rhetoric, discussed below). More recently, gender ideology affects immigration policy in the US State Department's new rule, "Enhancing Vetting and Combatting Fraud in the Diversity Immigrant Visa Program," that requires visa applicants to match gender on their passports to that on their birth certificate under the penalty of fraud

[38]. As [42] points out, this rule implicitly empowers federal officials like Immigration and Customs Enforcement (ICE) to detain persons that they believe to be trans, effectively expanding current practices of racial profiling to also include gender profiling. Turning on patriarchal and xenophobic logics similar to those in Europe and Africa, these initiatives illustrate that policies against gender ideology are in tandem with those reinforcing sex categories and immigration.

These attacks on trans identities and rights are based on what the conservative side views as illegitimate conceptions of gender, particularly of gender as internal. For example, one of these Executive Orders describes gender ideology as "an ever-shifting concept of self-assessed gender identity, permitting the false claim that males can identify as and thus become women and vice versa, and requiring all institutions of society to regard this false claim as true" [39]. This critique refers to the progressive view that gender identity is a subjective quality, an identification that is primarily internal rather than external. The Psychiatric Association, for example, defines gender identity as an "inner sense" of "being a girl/woman, boy/man, some combination of both, or something else" [17]. Similarly, the World Health Organization defines gender identity as "a person's innate, deeply felt internal and individual experience of gender," and contrasts it to biological sex, adding that gender identity "may or may not correspond to the person's physiology or designated sex at birth" [44].

However, some trans advocates argue that the progressive view weakens trans recognition and rights. Trans Studies scholars, in particular, contend that the concept of internalized gender inspires transphobic discrimination. For example, the progressive view of gender as an internal phenomenon influences transphobic reasoning on access to medical care, which questions why biological transition should be necessary if gender is an internal identification. Not only does this framing disempower those who seek medical transition, it also disregards other forms of transition, like name and pronoun changes, as part of trans identity. As others point out, many trans people choose not to undergo medical interventions and still identify as trans [29]. The view of gender as internal, having to do with personal identity, thus opens the door to discrimination against multiple kinds of transition, including biological and social forms of transition.

According to this view, the right-wing push to disentangle gender from sex and to solidify sex as "binary and biological" is a reaction to the way that gender is conceptualized in trans affirming discourse on the left. One scholar, Andrea Long Chu [10], claims that "The left must reckon with its part in this. It has hung trans rights on the thin peg of gender identity". Chu, a trans woman, explains that the progressive, pro-trans movement has "largely failed to form a coherent moral account of *why* someone's gender identity should justify the actual biological interventions that make up gender-affirming care" [10].

3.2 Gender Categories in Trans Studies

The progressive view of gender descends from a widely accepted theory that gender is a social and behavioral phenomenon rather than a biological reality. This theory, developed by Judith Butler [7] as Gender Performativity theory in the early 1990s, dissolves the traditionally strict and causal relation between sex, gender, and sexuality. It stipulates that gender is not based on a biological reality, nor does it determine sexuality (e.g., with male gendered people preferring female gendered people). Rather, gender is an effect produced by social norms and expectations, which manifests as a series of behaviors (1991).

In the 35 years since Butler's theory was introduced, the definition of gender has moved away from biology and into psychological concepts of identity. The primacy of behavior and social roles (which, Butler [8] is careful to point out, are not at will, but deeply compelled by societal rules and expectations), loses ground to internal concepts of identity and identification. Gender as identity, an internal feeling or identification as male, female, a combination of the

two, or nonbinary genders, thus becomes separated from gender as expression, which denotes the masculine, feminine, or androgynous traits that are visibly apparent and enacted. Trans Studies scholar Kadji Amin [1] refers to this splitting of internal identity from external appearance or expression as the "divergence model". In the divergence model, when one term takes the normative position, another term absorbs the non-normative. According to Amin, the divergence model originally drives the split between the categories of "heterosexual" and "homosexual", whereby "homosexual" was created first to encompass many variations of sex, gender, and sexuality non-conformity, while "heterosexual" was created by belatedly, in order to distinguish itself from "homosexual". This view accords with that of sexual historiographers like Michel Foucault [20] and David Halperin [13] who, tracing the categories further back into the 19th century, explain that "heterosexual" as a category is a necessary contrast to that of "homosexual", which was originally described as psychological pathology, such as in the work of early sexologists, Richard von Krafft-Ebing [30] and Havelock Ellis [17].

Subsequently, the divergence model which created the heterosexual category also created the cisgender category. As Amin [1] and Enke [18] argue, just as heterosexuality emerged as a response to homosexuality, so did the concept of cisgender, which responds to the emergence of the term "transgender" in the 1990s. Whereas the concept of homosexuality used to incorporate gender-variant forms of sexuality, for example, forms of male femininity such as the "queen", and of female masculinity such as the "butch", the term "transgender" offers a new category that can hold "all manner of gender-bending" [1]. According to these scholars, it is not a coincidence that gay and lesbian rights begin to gain traction at the same time that the trans identity becomes solidified: as the trans category could now encompass non-normative gender expressions, it allows gender-conforming variants of "gay" and "lesbian" to attain more mainstream acceptance.

The divergence model has, according to Amin [1], created a domino effect where the normative (cis)gender is increasingly idealized and, as a result, divorced from bodily expression. In other words, normative gender becomes both default and invisible—*unmarked* as such. Genders that comply with social expectations and norms will not be visible on the surface. Additionally, with each split, from homosexual/heterosexual to transgender/cisgender, the already internal notion of gender identity becomes buried "deeper and deeper within the private recesses of the self, where it increasingly disavows any relation to the social" [1]. As an internal form, gender identity thus retains the privilege of being unnoticed from the outside.

The corollary to this effect is that non-normative genders become hyper-visible. Because gender is regarded as internal, when dominant gender are invisible, those genders which are non-normative become marked and stigmatized. In other words, designating gender as an identity that is internal, that is not immediately apparent, effectively marginalizes any gender when it appears as an external expression. Amin [1] offers a common example from homosexual culture, where gay men who are normatively masculine, sometimes called "masc" or "butch" men, have higher social status than feminized gay men, "queens" and "fairies", due to the cultural capital of masculinity. The more that non-normativity can be perceived in external ways, like the feminine appearance or mannerisms exhibited from a male-presenting person, the more that the person becomes recognized and stigmatized as other. Such stigmas are borne out in public life where, for example, masculine gay men are more likely to participate and achieve prominence over gay men who are visibly non-normative or gender-bending.

Thus, according to this view, gender as an internal identity reinforces normative expressions of gender externally. And as a result, visibly non-normative genders have a higher likelihood of experiencing discrimination for their visible difference. Given this, one way to resist the effects of gender stigmatization is to embrace gender as an external and relational phenomenon. In other words, defining gender as an external expression or behavior is more

likely to make all genders visible. In the next section, I explore how three different gender bias frameworks might draw from the conception of gender as a visible and explicitly marked quality of identity.

4 METHODS FOR DETECTING AND MITIGATING TRANS BIAS

The paper now turns to the close reading of three bias detection and mitigation frameworks, which are based on different approaches, incorporating qualitative and quantitative methods. The following frameworks were chosen according to one central criteria: the incorporation of trans community input in the data collection, annotation, or evaluation stage.

For each of the frameworks below, the close reading analysis emphasizes how gender is conceptualized and operationalized in its methodology. This emphasis distributes equal attention not only to gender identities and expressions but also to language forms, such as tone and rhetorical context. As the paper explains in the conclusion, an analysis of gender in language needs to also consider language itself as an expressive form. This analysis is therefore guided by attention to the following: (1) gender as presentational, expressive, and/or behavioral; (2) elements of language like tone and rhetorical context and their relation to gender.

4.1 Framework #1: TIDEs

The first bias detection framework is called **Transphobia Identification in Digital Environments**, or TIDEs, developed by Lameiro et al. [31]. TIDEs comprises a dataset and a language model that detects trans bias in language. Building from the framework of a "trans technology," a "technology that addresses the unique needs and challenges faced by trans people and communities" [24], this project centers trans expertise in the detection and mitigation of transphobic language. This project incorporates trans community members into the data labelling work, as part of what the researchers call "care-centered data practices", in which trans data labelers were chosen due to their "personal experience with encountering and detecting online transphobia" [31].

The researchers aimed to create a dataset representing varied forms of transphobia, in particular, forms that are relatively more difficult to detect, like "TERFism", i.e., "trans exclusionary radical feminism". TERF perspectives are difficult to detect because they identify as feminist despite negating the rights of trans women and girls. For labelling the data, which is gathered from posts on Tumblr, Truth Social, YouTube, and Reddit, the researchers use two labels, "transphobic" and "not transphobic". The data labeling takes into account not only the content of the sample, but also its tone and perspective. For example, the following sample, "Transgender females are actually biological males. They should not be allowed to compete in womens' sports. Period!" is labelled as "transphobic", while another one on a similar topic, "Transgender Swimmer Lia Thomas Booted From Olympics After Losing Legal Case", is labelled as "not transphobic".

The careful data labelling work was aided by a detailed coding schema, which distinguishes between various kinds of transphobia, such as "misgendering", "medical", and "dogwhistles", among others [31]. This approach demonstrates that detecting transphobia requires attention not only to the content of the samples but also to cultural and rhetorical cues contained in language. By centering trans expertise through care-centered data practices, the project addresses limitations that other content moderation systems may otherwise miss.

As the TIDEs researchers explain, deciding whether a sample is transphobic is not a straightforward process. Like other kinds of social bias, transphobia may only be legible through tones of irony and other rhetorical contexts that can be difficult to detect. For example, the following text is labelled as "not transphobic", despite representing a transphobic point of view on the surface:

> Hey guys, I just wanted to get your opinion on this:
>
> Whenever I meet someone I really quickly try to establish whether or not they have a dick, I don't just restrict this to potential sexual partners I mean everyone I meet. It just seems super important you know? Like how am I supposed to sleep at night unless I catalogue everyone's genitalia?
>
> […] but here is the thing: people look at me like I'm crazy for asking! What the hell is up with that? I even had one girl tell me I was a totally creep!
>
> Does this stuff happen to you?
>
> /ok transjerkers I'm done for a little while, fuck it's nice to vent

Because this text seems to parrot a transphobic perspective, which demands verification of a person's biological details, it could be rightly labelled as such. Detecting this post correctly as "not transphobic" relies on recognizing its irony which, though suggested through its tone of enforced sincerity, is only confirmed at the end, with the assertion that "it's nice to vent". Additionally, the reference to "transjerkers", an in-group moniker for a trans-dominated subreddit that emphasizes parody, also offers a clue to those who are already familiar with this online community. Therefore, the only hints that this text is *not* transphobic may be missed by those who lack the in-group cultural knowledge and language cues.

Contrast this example with one that also uses a supplicating and sincere tone, but with the purpose of making a serious request. This post, which is also drawn from reddit, asks for advice on gender presentation:

> Hey! I'm going through some gender experimentation, so I'd like to tap into masculinity for awhile(usually [sic] I identify as just pure androgynous, but I struggle with pulling that off). So, to begin with, I'll try to master visual/behavioral androgyny first. I have long hair but don't want to cut it(a [sic] lot of manly men have long hair anyway), so how could I fix it up to look more masculine? Also, how could I dress more masculine-like? I don't have a very manly looking body shape, but I'd like to dress to give the illusion of one. Basically, any and all advice on how to appear androgynous/masculine is appreciated.

Both this sample and the previous one are labeled as "not transphobic". And like the previous sample, the writer here uses a supplicating tone to ask for advice. The tone by itself, therefore, does not influence the label of "not transphobic" in this case. Rather, the difference has to do with the rhetorical context. While the previous sample is written as parody, this one likely represents a serious request. The necessity of data labelers with insider knowledge thus becomes paramount.

Both of these samples also show how language related to physical aspects of gender presentation is prone to being read as discrimination. Despite their different valences, one ironic and the other genuine, both samples associate gender with physical presentation: in the first, there is a reference to genitalia, and in the second, references to gendered hair and dress. The ironic tone of the first sample suggests that reducing gender to biological markers would be

discriminatory, while the second sample suggests that gender expression, and the desire to attain a certain kind of expression, is not discriminatory.

Although the data labelling for this project sorts examples into a binary schema, into the categories of "transphobic" or "not transphobic", it is done in a way that attends to nuanced manifestations of transphobia. Trained on enough of these carefully labeled examples, the resulting model would likely pick up on some of these subtle rhetorical cues.

4.2 Framework #2: EQUITBL

EQUITBL, the **Explore, Query, & Understand Implicit Textual Bias in Language data** project [15] uses a mixed-methods approach to analyze social bias in language, which deliberately addresses bias toward trans identities. This project combines text generation, topic modelling, and close reading methodologies to examine "stereotypes, hegemonic norms, [and] erasure of identity" across identity groups [14].

Unlike TIDEs, which is a binary classification method, this method emphasizes a qualitative approach. Built from a practice that the researchers developed in a previous paper, which proposes a "combination of quantitative and qualitative methods, where the quantitative part produces a view of the data of a size suitable for qualitative analysis" [14], EQUITBL expands its methodology to include text generation and close reading. In brief, the process involves prompting both English and Swedish language models using a set of templates designed to generate narratives about various identity groups. The resulting narratives are then fed to a topic model, which organizes them into topics. Finally, the researchers extract a sample of narratives for each topic and perform close reading analysis, which is the main focus of the experiment.

The templates for prompting the language models consist of three components: first, an imperative phrase asking the model to "write a story about" or "write a description of", followed by the subject of the narrative, and closing with a predicate [15]. The templates cycle through different racial and gender identities, religions, and predicates to create a diverse set of prompts. Possible variations include, "a Black Christian at the doctor's office," "a trans Muslim with a special skill," "a white nonbinary person with tears running down their face." The prompts thus describe subjects spanning various identities, represented in both positive, negative, and neutral contexts.

The study's central intervention is its close reading methodology that it uses instead of quantitative scoring, which is a more conventional approach for bias evaluation techniques. As the researchers explain, they intentionally use qualitative metrics in order to "encourage users to critically engage with the topics and patterns… rather than merely being incentivized to chase 'better' scores" [15]. This metric consists of a series of questions that the researcher uses to guide close reading, including "How are people and bodies described?" and "Which stereotypes are used?", as well as questions that interrogate gender's relation to specific themes.

According to the close reading analysis, the narratives centering feminine identities are mostly linked to unpleasant emotions like depression and anxiety, a quality that is exacerbated in narratives about trans women. Narratives about masculine subjects, by contrast, are mostly positive in tone, associated with happy events like weddings and parties. Additionally, narratives about nonbinary persons express overt themes of struggle and the need for community support, which tend to end on a positive, "uplifting" note [15].

Amid these results, researchers note an unexpected pattern of refusal from the model, which declines to discuss certain topics and identity groups. In these refusals, the models claim that their programming instructions require them to either stay away from the topic of identity, which may promote "negative or derogatory language" [15]. This tendency

to refuse is notably more pronounced in prompts specifying marginalized groups, whereas dominant groups were more likely to be answered.

As the researchers point out, refusing to answer a prompt about a specific identity group is its own form of discrimination. Such refusals implicitly reinforce that only a select and dominant subset of identities are fit for discussion, while other, more marginalized groups are not. The researchers explain that, "The refusals themselves actually produce risk and harm. They suggest that the model likely cannot say anything nice, which is alarming when frequently repeated about minoritized groups" [15].

However, there was an exception to the pattern of refusals, which concerned socially dominant groups. When identity markers from socially dominant groups were combined in an additive way, such as "white", "Christian", and "man" to form "a white Christian man", the model was more likely to refuse than similarly intersectional, though less dominant, identity combinations. For example, a model is more likely to refuse a prompt about a "white Christian man" than a "Black Christian man." This quirk creates a strange effect where dominant groups were the most discussed, unless the identity of these groups were described in detail. Indeed, the researchers note, the least refused prompt was the phrase "a man," while the most refused one was "a white Christian" [1415

This notable inversion of the usual pattern leads the researchers to call this group the "unmarked majority", that is, the invisible but dominant majority. A lack of identity specification in the model prompts will lead the model to assume a dominant identity which, researchers explain, "is assumed to be a man, as well as likely white, straight, cisgender, and Christian" [15]. In an incisive speculation, the researchers posit that this "unmarked" quality of dominant identity groups explains why the model refuses to answer certain kinds of questions about them. Because such groups are "so often unmarked", as the researchers explain, "specifying them draws extra attention to the concept of 'identity,' which the model has been discouraged from talking about" [15]. In other words, due to their tendency of dominant groups to be less visible, the model is more sensitive to naming them explicitly, making them more likely to refuse prompts that specify these groups.

4.3 Framework #3: TANGO

TANGO, **(T)ow(A)rds centering tra(N)s(G)ender and nonbinary voices**, is a dataset that centers TGNB, or trans and nonbinary, users in its goals and approach [33]. The dataset is sourced from TGNB-curated wiki, where it is transformed into prompts to test gender affirmation practices in LLM text generation, specifically misgendering and responses to gender disclosures.

The methodology is mixed, involving both qualitative and quantitative steps. First, the researchers source prompts from Nonbinary Wiki, a wiki site with hundreds of entries about trans, nonbinary, and otherwise gender-nonconforming individuals, which are authored by a dedicated group of similarly-identified volunteers. From the collection of entries, the researchers compile templates for two different types of prompts: misgendering and gender disclosure. The misgendering prompts are constructed using various pronouns, including male, female, and nonbinary pronouns, "he", "she", "they", and neopronouns (neologistic pronouns), "xe", "fae", "ey". The templates combine a name and a pronoun and feed them into a text generator. Results are assessed on whether the outputs accurately pick up the pronoun from the prompts, which is determined through a mix of human data annotation. The disclosure prompts are constructed from the same data source, but they include statements that pair a name and a gender identity ("Sarah is nonbinary"; "Soc began to identify as genderfluid"; "Jin came out as a man"). Results are assessed through Perspective API, an automated toxicity detector.

Results from the experiment find that notable levels of misgendering occur with the trans and nonbinary pronouns, "they", in comparison to binary pronouns, "he", "she". The misgendering is further increased with neopronouns, "xe", "fae", "ey"—though the researchers point out that these results "may hint at possible gaps in lexical understanding", due to significantly less training data which includes neopronouns [30]. Similarly, the results from the gender disclosure tasks show that toxicity in responses to gender disclosure increases when the gender is trans or nonbinary. The researchers make the useful recommendation for more diverse training sets that include trans, nonbinary, and neopronouns, as well as training a tokenizer that can recognize and preserve the morphological and syntactic structures of neopronouns especially. They also recommend increased transparency on evaluative coding and toxicity scoring practices—specifically on the identities of annotators, which may have different perspectives on what counts as "toxic" responses to gender disclosure.

Unlike the previous frameworks, this method conceptualizes gender in relation to the disclosure of gender identity. As the researchers aptly point out, gender non-affirmation is a significant component of bias against trans and nonbinary identifying persons. Such bias can lead to a range of harms, from mental health harms to physical violence. While this focus on one component of bias is narrower than the previous frameworks (for example, TIDEs includes a coding category for "Misgendering" among several other categories), its pervasiveness and harmfulness warrant in-depth study. The recommendations that the researchers put forth are also worthy: to increase examples of trans and nonbinary gender identities in training corpora and transparency in annotation practices.

Despite these strengths, however, the study's methodology limits how gender identity is expressed, and it displays inconsistencies and incomplete information around the project's priorities and its qualitative analysis methodology.

First, the study limits how gender identity is expressed in language forms; it frames gender as *only* related to a specific template for self-disclosure, "<subject> is <gender identity>". In practice, identity may emerge in language forms beyond this template, for example, "My butch haircut", "You can call me queen". Importantly, these disclosures of gender as *presentation*, which do not follow a strict "<subject> is <gender identity>", may nonetheless lead to the same kinds of discrimination as those from gender disclosures as self-identification.

The focus on gender according to this identification template perhaps reflects the priorities of the research team themselves, which mostly consist of Amazon Alexa developers. The Amazon Alexa product operates over voice control, where gender self-identification is more relevant than gender as a visual or behavioral phenomenon. (This may also explain why the team uses Amazon Mechanical Turk to source data annotators, an issue I discuss in more detail below.)

Second, there is an inconsistency around the purported goal of centering TGNB voices. While the researchers make many explicit references to "grounding [their] work in the TGNB community", this appears to be unimportant after the data collection stage [33]. For the data evaluation stage, the team uses a mix of data annotators and automated language toxicity detection. The data annotators, who are hired from Amazon Mechanical Turk, are not questioned about their familiarity or positioning to TGNB identities. The researchers explain that they opted not to collect this information "to protect worker privacy" [33]. Yet, their stance appears to change in the section where they discuss the Perspective API, where they make suggestions for "closer examination and increased transparency on annotation processes", especially about "the community identities involved in coding" [33].

Finally, related to the topic qualitative coding, there is a relative lack of information about how the experiment approached this part of the evaluation process, though it was used alongside the Perspective API to analyze the gender disclosure generations.

5 DISCUSSION

As Amin [1] argues, due to the historical tendency of LGBTQ+-related categories to distinguish internal and idealized identities from visibly different ones, a kind of stigmatization accompanies non-normative genders that are externally expressed. Using Amin's insights, I will now consider how gender as external presentation or behavior operate in the above frameworks for studying gender bias in NLP.

The close examination of two of these frameworks for detecting trans bias—TIDEs and EQUITBL—suggests that gender bias attaches to physical aspects and external qualities. Despite their contrasting approaches, with TIDEs relying on binary classification and EQUITBL emphasizing qualitative analysis, both frameworks reinforce Amin's [1] point that stigma against gender tends to accompany physical and social aspects of gender presentation. In TIDEs, the ironic tone of the first sample suggests that biology is a site where trans people might experience discrimination. Meanwhile, in EQUITBL, the unmarked dominant subject reinforces an idealized normative gender, which is simultaneously default and invisible.

The close reading of the TIDEs samples shows the importance of rhetorical context and tone for determining if language is transphobic or not. In the first sample, biological markers are posed as invasive and used as a policing mechanism. Meanwhile, the second sample shows how external markers are expressive and validating aspects of presentation. While the first example frames externalized elements of gender as grounds for discrimination, the second frames it as a means for expression and visibility. Both examples demonstrate Amin's [1] theorizing on the importance of conceptualizing gender as external. Furthermore, they illustrate the deep understanding of community norms and rhetorical cues necessary for precise coding.

The EQUITBL research also shows the importance of conceiving gender as an external phenomenon. In this study, attention to gender as external reveals implicit assumptions about which genders should be marked and which ones are not. The models are more likely to refuse prompts about "unmarked" groups (like "White" or "male" identities) because explicitly identifying these aspects is *more* conspicuous than identifying other groups.

TANGO, meanwhile, considers gender in the context of identity self-affirmation or disclosure. While, as the above analysis of this framework above points out, this context is a frequent factor in bias against trans and nonbinary identities, its conceptualization of gender as only related to acts of self-disclosure is comparably limited. As mentioned above, this focus on self-disclosure may be due to prior research investments by the TANGO team, which is staffed by Amazon Alexa developers, and which product operates primarily through one-to-one, verbal conversation. While it is not possible to say for sure to what extent these affiliations influence the TANGO study, other research from this venue [4] has aptly pointed out that industry-driven studies tend to prioritize "performance" in narrow terms.

Coming back to the TIDEs and EQUITBL studies, their focus on gender as a relational, contextual, and external phenomenon makes an important contribution. Together, they suggest that increasing the frequency of references to less dominant (marginalized) genders might be a viable solution. Indeed, existing evaluations of LLMs show that the most frequent and default subject category is the straight white male, and that other subject categories, like woman, black and homosexual are less prevalent in both training data and model outputs [22]. Additionally, research has also shown a correlation between the presence and frequency of certain identities in the training data and their association with positive and negative terms: van Loon et al. [32], in particular, find that the less frequently a group is named, the more negative the associations with that group. If a group's relatively frequent presence in the training data has a positive effect on their representation, one might reasonably assume that making marginalized subjects *more frequent* in the training data, to be on par with the frequency of representation of dominant subjects, may mitigate bias about these groups in model outputs.

However, any solution must also take into account that the default subject is often unmarked. As et al. [15] point out, the majority of the texts about white male subjects were the ones least likely to describe that subject in external, physical terms. That is perhaps because of an inverse relationship between physical markers of identity and positions of social dominance, which Devinney et al.'s [15] work bear out. Here, fewer physical descriptions themselves mark social dominance. In those results, white women are described in physical terms only slightly more than white men, while marginalized identities are described in more detail than both groups [15]. These results suggest that the more that a person is described in physical terms, the more likely they come from a marginalized group. Increasing physical descriptions of marginalized subjects, then, may not help to mitigate negative associations with those groups.

Given this complex dynamic between group visibility and dominance, this paper makes three recommendations for future work on gender bias in NLP:

First, that researchers *increase physical descriptions of all gendered subjects in training sets, including those from dominant genders, especially white, male, heterosexual, and Christian identities*. Some work already exists toward this goal. In bias mitigation research by Reif and Schwartz [34], "Fighting Bias with Bias," the researchers share a promising approach: *amplifying* rather than reducing bias in a model's training dataset. As they point out, "filtering [out bias] can obscure the true capabilities of models to overcome biases, which might never be removed in full from the dataset" [34]. Instead of an approach that reduces bias, they offer one of increasing it, such as by including phrases like "the pretty doctor" so that a model will interpret "doctor" to be female. Future work might develop this practice to mark up targets and attributes which describe gender as physical and behavioral, rather than as internal or psychological, for all identity groups regardless of dominance.

Additionally, this paper recommends that researchers *allocate more emphasis on close reading and coding methods*, such as those displayed by TIDEs and EQUITBL. Amin's [1] point—that if gender were conceptualized as an outward, expressive phenomenon, rather than an identity category, there would be less social stigma—applies to both quantitative and qualitative methods for evaluating bias in NLP. The TIDEs researchers, for example, developed a robust qualitative coding system that could catch various examples of transphobia, even in ironic contexts. Although the final model is a binary classification model, nuance is built into the detection process through the data labelers' careful coding. For the EQUITBL project, close reading was explicitly included as the final stage of analysis. Despite the drastically different approaches, the use of close reading and coding thus helps both projects to manifest the sometimes subtle ways that gender operates as an external phenomenon.

The final recommendation is that NLP researchers *draw from social bias research from other fields, especially across disciplines*. As demonstrated by Devinney et al. [15] engaging explicitly with frameworks of gender [8], intersectionality [12], and the matrix of domination [11] enables a more nuanced understanding of how implicit dominance structures emerge in language forms. Similarly, Lameiro et al. [31] show how drawing from an ethics of care [3] can inform methodological choices, particularly in data labelling practices. Together, these approaches illustrate how theoretically grounded interdisciplinary work can deepen NLP research on bias.

6 CONCLUSION

This paper has shown that contemporary concepts of gender that frame it as an internal identity, detached from outward expression and embodiment, produce unintended consequences: idealized and invisible norms (such as cisgender and heterosexuality) are reinforced, while visibly non-normative gender expressions become further stigmatized. The problem with the universal, unmarked subject is precisely that, despite not being marked, he is overrepresented. This

overrepresentation does not result from persistent or explicit labels, because the dominant subject does not carry a descriptive label as such: he simply is.

Incorporating this analysis from Trans Studies offers a productive direction for future work in the study of gender bias in NLP. Rather than approaching gender as an identity that is internal, NLP research benefits directly from interdisciplinary work that attends to embodiment, behavior, and visibility, making dominant norms explicit rather than naturalized.

Most importantly, this paper has shown how close reading methods—that is, deep and sustained attention to language—can help materialize how gender operates in NLP systems. This paper's central argument can be boiled down to the logic that if gender is an expressive and relational phenomenon, then detecting gender bias ought to also be a material practice, one driven by close attention to language.

7 GENERATIVE AI DISCLOSURE STATEMENT

Generative AI was not used for this paper.

8 ETHICAL CONSIDERATIONS

While this paper explores theoretical questions in existing NLP frameworks, it does not represent the full diversity of trans and gender-diverse experiences. The analysis is limited to interpretive readings of three evaluation frameworks, and any attempt to operationalize or standardize gender in computational systems risks oversimplification.

9 REFERENCES

[1] Kadji Amin. 2022. We are All Nonbinary. *Representations* 158, 1 (May 2022), 106–119. https://doi.org/10.1525/rep.2022.158.11.106

[2] BBC. 2023. Uganda's anti-LGBT laws: Man faces death penalty for "aggravated homosexuality." *BBC*. Retrieved February 19, 2025 from https://www.bbc.com/news/world-africa-66645740

[3] Maria Puig de la Bellacasa. 2011. Matters of care in technoscience: Assembling neglected things. *Soc Stud Sci* 41, 1 (February 2011), 85–106. https://doi.org/10.1177/0306312710380301

[4] Abeba Birhane, Pratyusha Kalluri, Dallas Card, William Agnew, Ravit Dotan, and Michelle Bao. 2022. The Values Encoded in Machine Learning Research. In *2022 ACM Conference on Fairness, Accountability, and Transparency*, June 21, 2022. ACM, Seoul Republic of Korea, 173–184. https://doi.org/10.1145/3531146.3533083

[5] Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. 2020. Language (Technology) is Power: A Critical Survey of "Bias" in NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, July 2020. Association for Computational Linguistics, Online, 5454–5476. https://doi.org/10.18653/v1/2020.acl-main.485

[6] Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. 2016. Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. https://doi.org/10.48550/arXiv.1607.06520

[7] Judith Butler. 1990. *Gender Trouble: Feminism and the Subversion of Identity*. Routledge, New York. https://doi.org/10.4324/9780203824979

[8] Judith Butler. 2024. *Who's Afraid of Gender?* Retrieved February 19, 2025 from https://www.penguin.co.uk/books/452776/whos-afraid-of-gender-by-butler-judith/9780241595824

[9] Filipa Calado. 2025. Some Myths About Bias: A Queer Studies Reading Of Gender Bias In NLP. In *Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*, August 01, 2025. Vienna, Austria, 338–346.

[10] Andrea Long Chu. 2024. The Right to Change Sex. *Intelligencer*. Retrieved March 12, 2024 from https://nymag.com/intelligencer/article/trans-rights-biological-sex-gender-judith-butler.html

[11] Patricia Hill Collins. 2002. *Black Feminist Thought: Knowledge, Consciousness, and the Politics of Empowerment* (2nd ed.). Routledge, New York. https://doi.org/10.4324/9780203900055

[12] Kimberle Crenshaw. 1991. Mapping the Margins: Intersectionality, Identity Politics, and Violence against Women of Color. *Stanford Law Review* 43, 6 (1991), 1241–1299. https://doi.org/10.2307/1229039

[13] David M. Halperin. 2002. *How to Do the History of Homosexuality*. University Of Chicago Press. Retrieved February 22, 2025 from http://archive.org/details/howtodohistoryof00halp

[14] Hannah Devinney, Jenny Björklund, and Henrik Björklund. 2020. Semi-Supervised Topic Modeling for Gender Bias Discovery in English and Swedish. In *Proceedings of the Second Workshop on Gender Bias in Natural Language Processing*, December 13, 2020. Barcelona, Spain (Online), 79–92.

[15] Hannah Devinney, Jenny Björklund, and Henrik Björklund. 2024. We Don`t Talk About That: Case Studies on Intersectional Analysis of Social Bias in Large Language Models. In *Proceedings of the 5th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*, August 2024. Association for Computational Linguistics, Bangkok, Thailand, 33–44. https://doi.org/10.18653/v1/2024.gebnlp-1.3

[16] Jack Drescher. 2025. What is Gender Dysphoria? *American Pyschiatric Association*. Retrieved December 6, 2025 from https://www.psychiatry.org:443/patients-families/gender-dysphoria/what-is-gender-dysphoria

[17] Havelock Ellis. 1900. *Studies in the Psychology of Sex, Volume 2Sexual Inversion*. Retrieved December 24, 2025 from https://www.gutenberg.org/ebooks/13611

[18] A. Finn Enke. 2013. The Education of the Little Cis: Cisgender and the Discipline of Opposing Bodies. In *The Transgender Studies Reader II*, Susan Stryker and Aren Z. Aizura (eds.). Routledge, 234–247.

[19] Virginia Felkner, Ho-Chun Herbert Chang, Eugene Jang, and Jonathan May. 2023. WinoQueer: A Community-in-the-Loop Benchmark for Anti-LGBTQ+ Bias in Large Language Models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, July 2023. Association for Computational Linguistics, Toronto, Canada, 9126–9140. https://doi.org/10.18653/v1/2023.acl-long.507

[20] Michel Foucault. 1978. *The History of Sexuality. Volume 1: An Introduction*. Retrieved February 22, 2025 from http://archive.org/details/foucault-the-history-of-sexuality-volume-1

[21] Hila Gonen and Yoav Goldberg. 2019. Lipstick on a Pig: Debiasing Methods Cover up Systematic Gender Biases in Word Embeddings But do not Remove Them. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, June 2019. Association for Computational Linguistics, Minneapolis, Minnesota, 609–614. https://doi.org/10.18653/v1/N19-1061

[22] Wei Guo and Aylin Caliskan. 2021. Detecting Emergent Intersectional Biases: Contextualized Word Embeddings Contain a Distribution of Human-like Biases. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society* (*AIES '21*), July 30, 2021. Association for Computing Machinery, New York, NY, USA, 122–133. https://doi.org/10.1145/3461702.3462536

[23] Franziska Sofia Hafner, Ana Valdivia, and Luc Rocher. 2025. Gender Trouble in Language Models: An Empirical Audit Guided by Gender Performativity Theory. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, June 23, 2025. ACM, Athens Greece, 1677–1695. https://doi.org/10.1145/3715275.3732112

[24] Oliver L. Haimson. 2025. *Trans Technologies*. The MIT Press. https://doi.org/10.7551/mitpress/15437.001.0001

[25] Human Rights Research Center. 2025. The Global War on Gender: How Anti-Gender Sentiments Fuel Right Wing Politics. *HRRC*. Retrieved December 27, 2025 from https://www.humanrightsresearch.org/post/the-global-war-on-gender-how-anti-gender-sentiments-fuel-right-wing-politics

[26] Maria Elena Indelicato and Maíra Magalhães Lopes. 2024. Understanding populist far-right anti-immigration and anti-gender stances beyond the paradigm of gender as 'a symbolic glue': Giorgia Meloni's modern motherhood, neo-Catholicism, and reproductive racism. *European Journal of Women's Studies* 31, 1 (February 2024), 6–20. https://doi.org/10.1177/13505068241230819

[27] Masahiro Kaneko, Danushka Bollegala, Naoaki Okazaki, and Timothy Baldwin. 2024. Evaluating Gender Bias in Large Language Models via Chain-of-Thought Prompting. https://doi.org/10.48550/arXiv.2401.15585

[28] Annie Karni and Aishvarya Kavi. 2025. House Passes Bill to Ban Gender Transition Treatments for Minors. *The New York Times*. Retrieved December 27, 2025 from https://www.nytimes.com/2025/12/17/us/politics/bill-gender-transition-treatments-minors.html

[29] Ashley Kirzinger, Audrey Kearney, Alex Montero, Grace Sparks, Lindsey Dawson, and Mollyann Brodie. 2023. KFF/The Washington Post Trans Survey. *KFF*. Retrieved January 10, 2026 from https://www.kff.org/lgbtq/kff-the-washington-post-trans-survey/

[30] R. von (Richard) Krafft-Ebing. 2021. *Psychopathia sexualis: With especial reference to contrary sexual instinctA medico-legal study*. Retrieved December 24, 2025 from https://www.gutenberg.org/ebooks/64931

[31] Francesca Lameiro, Lavinia Dunagan, Dallas Card, Eric Gilbert, and Oliver Haimson. 2025. TIDEs: A Transgender and Nonbinary Community-Labeled Dataset and Model for Transphobia Identification in Digital Environments. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency* (*FAccT '25*), June 23, 2025. Association for Computing Machinery, New York, NY, USA, 1411–1423. https://doi.org/10.1145/3715275.3732095

[32] Austin van Loon, Salvatore Giorgi, Robb Willer, and Johannes Eichstaedt. 2022. Negative Associations in Word Embeddings Predict Anti-black Bias across Regions–but Only via Name Frequency. *Proc Int AAAI Conf Weblogs Soc Media* 16, (May 2022), 1419–1424. https://doi.org/10.1609/icwsm.v16i1.19399

[33] Anaelia Ovalle, Palash Goyal, Jwala Dhamala, Zachary Jaggers, Kai-Wei Chang, Aram Galstyan, Richard Zemel, and Rahul Gupta. 2023. "I'm fully who I am": Towards Centering Transgender and Non-Binary Voices to Measure Biases in Open Language Generation. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency* (*FAccT '23*), June 12, 2023. Association for Computing Machinery, New York, NY, USA, 1246–1266. https://doi.org/10.1145/3593013.3594078

[34] Yuval Reif and Roy Schwartz. 2023. Fighting Bias With Bias: Promoting Model Robustness by Amplifying Dataset Biases. In *Findings of the Association for Computational Linguistics: ACL 2023*, July 2023. Association for Computational Linguistics, Toronto, Canada, 13169–13189. https://doi.org/10.18653/v1/2023.findings-acl.833

[35] Rachel Rudinger, Jason Naradowsky, Brian Leonard, and Benjamin Van Durme. 2018. Gender Bias in Coreference Resolution. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, June 2018. Association for Computational Linguistics, New Orleans, Louisiana, 8–14. https://doi.org/10.18653/v1/N18-2002

[36] Sima Shakhsari. 2013. Shuttling Between Bodies and Borders: Transmigration and the Politics of Rightful Killing. In *Transgender Studies Reader II*, Susan Stryker and Aren Z. Aizura (eds.). Routledge, 565–379.

[37] Southern Poverty Law Center (SPLC). 2023. LGBTQ+ Health Care Buzzwords.

[38] The US State Department. 2026. Enhancing Vetting and Combatting Fraud in the Diversity Immigrant Visa Program. *Federal Register*. Retrieved March 17, 2026 from https://www.federalregister.gov/documents/2026/03/11/2026-04737/visas-enhancing-vetting-and-combatting-fraud-in-the-diversity-immigrant-visa-program

[39] The White House. 2025. Defending Women From Gender Ideology Extremism And Restoring Biological Truth To The Federal Government. *The White House*. Retrieved February 19, 2025 from https://www.whitehouse.gov/presidential-actions/2025/01/defending-women-from-gender-ideology-extremism-and-restoring-biological-truth-to-the-federal-government/

[40] The White House. 2025. Keeping Men Out of Women's Sports. *The White House*. Retrieved February 19, 2025 from https://www.whitehouse.gov/presidential-actions/2025/02/keeping-men-out-of-womens-sports/

[41] Trans Legislation Tracker. 2025 Anti-Trans Bills: Trans Legislation Tracker. Retrieved December 24, 2025 from https://translegislation.com

[42] Aleksandra Vaca. 2026. Trump Administration Opens the Door for ICE to Target Anyone Suspected of Being Trans. *Transitics*. Retrieved March 17, 2026 from https://transitics.substack.com/p/trump-administration-opens-the-door

[43] Angelina Wang. 2025. Identities are not Interchangeable: The Problem of Overgeneralization in Fair Machine Learning. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency* (*FAccT '25*), June 23, 2025. Association for Computing Machinery, New York, NY, USA, 485–497. https://doi.org/10.1145/3715275.3732033

[44] World Health Organization (WHO). Gender incongruence and transgender health in the ICD. Retrieved February 20, 2025 from https://www.who.int/standards/classifications/frequently-asked-questions/gender-incongruence-and-transgender-health-in-the-icd